

Wildfire containment probability is not affected by eastern spruce budworm defoliation in Ontario, Canada

Kennedy Korkola^{A,*}, Jennifer L. Beverly^B  and Patrick M. A. James^A

For full list of author affiliations and declarations see end of paper

*Correspondence to:

Kennedy Korkola
 Institute of Forestry and Conservation, John
 H. Daniels Faculty of Architecture,
 Landscape and Design, University of
 Toronto, Toronto, ON M5S 3B3, Canada
 Email: kennedy.korkola@mail.utoronto.ca

Received: 18 March 2025
Accepted: 24 July 2025
Published: 5 September 2025

Cite this: Korkola K *et al.* (2025) Wildfire containment probability is not affected by eastern spruce budworm defoliation in Ontario, Canada. *International Journal of Wildland Fire* **34**, WF25058. doi:10.1071/WF25058

© 2025 The Author(s) (or their employer(s)). Published by CSIRO Publishing on behalf of IAWF.

This is an open access article distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND)

OPEN ACCESS

ABSTRACT

Background. Stand-replacing wildfires and eastern spruce budworm outbreaks (*Choristoneura fumiferana*; SBW) are important disturbances in the boreal forest. SBW defoliation can affect fire behaviour by altering fuel loads and connectivity, thereby promoting the transition of low-activity surface fires into crown fires. However, little is known about how these altered fuels impact the effectiveness of fire suppression. **Aims.** To assess key drivers of initial attack (IA) success in Ontario's boreal forest and determine if incorporating SBW defoliation data improves predictive models. **Methods.** We developed random forest models of fire containment using established predictors including fire weather, fire size at IA and region. We then evaluated if the inclusion of time since SBW defoliation improved model performance. **Key results.** Fire size at IA was the most influential variable for determining whether a fire escaped containment. Contrary to our hypothesis, we did not find evidence that SBW defoliation greatly improved model performance. **Conclusions and implications.** The size of the fire at IA was the most important variable in determining successful containment. Although budworm defoliation has been shown to affect other aspects of fire hazard, we were unable to identify an influence on IA success. Future work could benefit from focused investigation into how historical SBW defoliation affects fire behaviour.

Keywords: containment probability, defoliation, fire containment, forest fire, initial attack success, machine learning, random forest, spruce budworm, wildland fire.

Introduction

The boreal forest is the most expansive forest type in Canada, providing numerous ecosystem services to local communities and the global population (Gauthier *et al.* 2015). The composition and configuration of the boreal forest that give rise to these services are shaped by dynamic landscape processes such as disturbances and succession, as well as topography, climate, and their spatial and temporal interactions (Turner 1989, 2005). Boreal ecosystems have evolved with natural disturbances such as wildfires (Weber and Stocks 1998) and periodic insect outbreaks (Blais 1983). However, disturbance regimes are shifting in terms of their frequency, severity and extent owing to climate change (Gauthier *et al.* 2015). These changes can produce novel and unexpected interactions that can alter landscape patterns and processes and thus shift ecosystems into alternative stable states (Dale *et al.* 2001; Buma 2015; Walker *et al.* 2023).

The eastern spruce budworm (*Choristoneura fumiferana*; SBW) is a species of native moth that defoliates large areas of balsam fir (*Abies balsamea*) and spruce (*Picea* spp.) trees throughout the central and southern Boreal, the Acadian and the northern Great Lakes–Saint Lawrence forest regions (Fig. 1a). Outbreaks occur periodically every 30–35 years and can last anywhere from 5 to 15 years (MacLean 1980; Williams and Liebhold 2000; Jardon *et al.* 2003), depending on forest composition, stand age (Blais 1983) and climatic conditions (Gray 2008). Forest stand mortality can occur by the fourth year of an outbreak (Morris 1963), but changes to forest structure can continue for up to 14 years (Watt *et al.* 2018).

These changes to forest structure following outbreaks have been hypothesised to influence subsequent disturbances such as wildfires (Stocks 1987). Fire behaviour is driven by

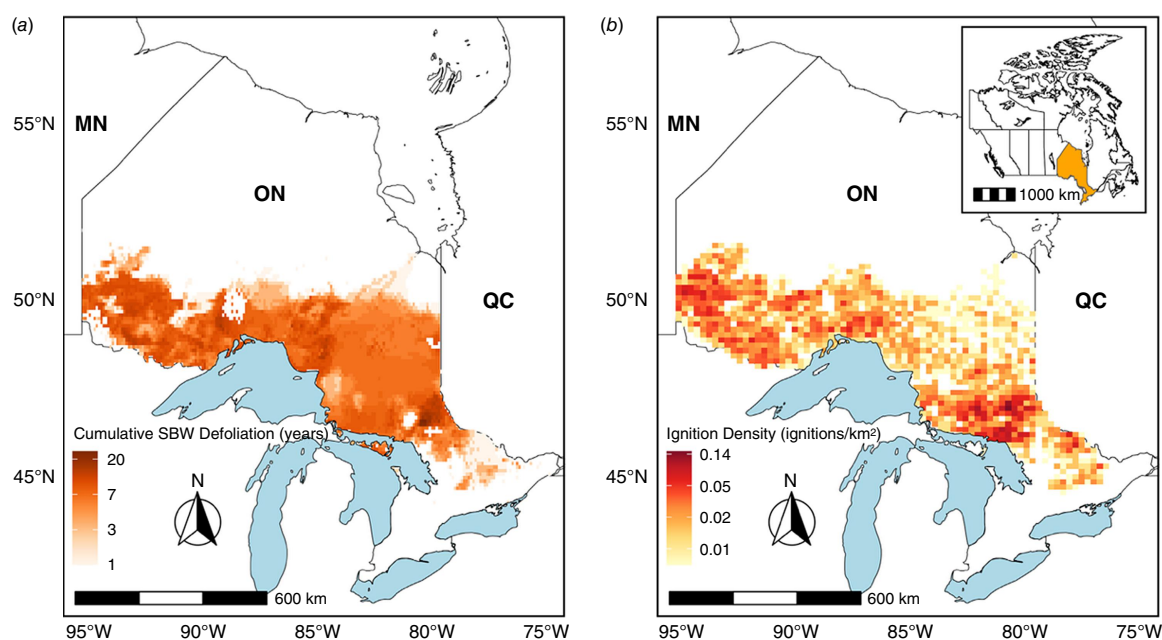


Fig. 1. The study region included the boreal shield of Ontario, Canada. Defoliation by the eastern spruce budworm (SBW) shares considerable spatial extent with fire ignitions across the province: (a) shows the area defoliated by the SBW between 1975 and 2018 shown as cumulative SBW defoliation in years, and (b) shows ignition density in the province of Ontario for 1990 to 2019 in ignitions per kilometre squared.

what is commonly known as the ‘fire behaviour triangle’, composed of three interacting elements: weather, fuels and topography (Countryman 1966). Changes to the forest structure directly affect fire behaviour by modifying the amount (i.e. load), connectivity, and moisture of the fuel. Increased surface fuel loads can create higher-intensity fires and greater fuel continuity from surface to crown that promote vertical fire spread (Watt et al. 2020) and high-intensity crown fires (Fleming et al. 2002; Candau et al. 2018). When stands have high canopy fuel loads and crown connectivity, active crown fires are more likely (Wagner 1977).

SBW defoliation can influence fire behaviour by altering forest fuel abundance, connectivity and structure at varying temporal and spatial scales (Watt et al. 2018). Cumulative defoliation has been positively correlated with vertical fuel connectivity (Watt et al. 2018) whereas the time since defoliation is associated with increased crown breakage and surface fuel accumulation (Stocks 1987; Watt et al. 2020). Defoliation begins at the tops of the trees, and following crown defoliation, the budworm begins feeding on older foliage, which ultimately leads to complete tree mortality (MacLean 1984). Dead trees, branches and needles accumulate owing to windthrow and breakage (Stocks 1987; Péché 1993; Watt et al. 2018), which contribute to the ladder fuels, lowering the crown to base height (CBH) and promoting fire spread into the canopy (Fleming et al. 2002). This change in fuel characteristics occurs at the hypothesised ‘window of opportunity’ (Fleming et al. 2002; Watt et al. 2020), a time lag of 3–9 years following defoliation in which fuel changes pose the most significant fire

risk (Fleming et al. 2002). Stocks (1987) found that within this timeframe, crown breakage went from 5 to 70%, thus increasing the surface fuel load. Watt et al. (2020) found a similar result, in which crown breakage increased by 200% within 4 years (Watt et al. 2020).

Models using historical data have also shown that cumulative defoliation affects the likelihood of lightning fire ignition in Ontario, where ignition probability increased following a lag of 8–10 years, hypothesised to be due to an increase in the surface fuel load (James et al. 2017). However, other studies looking at eastern SBW (Péché 1993) have found insect-related changes in fuels were not related to increased fire risk or crowning potential owing to high decomposition rates of downed wood (Péché 1993). A study conducted in the Cape Breton Highlands, a moist maritime climate, found there was no accumulation of fine fuels on the forest floor, as small woody fuels were fully incorporated into the forest floor within 1–2 years after falling to the surface (Péché 1993). Larger downed woody debris that did not decompose in the study time frame was too moist to contribute to fire propagation (Péché 1993). This dampening effect on fire was further exacerbated by the proliferation of the understorey vegetation, which reduced radiative heat transfer during a fire. Even under drought-like conditions, conventional initial attack (IA) tactics would be sufficient for wildfire suppression in moist maritime climates (Péché 1993). Other defoliators, like the western SBW (*Choristoneura occidentalis*), have been shown to decrease or have no effects on wildfire behaviour. Although this insect still affects fuel structure, it is hypothesised that fire

intensity (Cohn *et al.* 2014), occurrence (Lynch and Moorcroft 2008) and severity (i.e. ecological impact; Meigs *et al.* 2015) may be decreased owing to a reduction in canopy fuels and increased understorey regeneration following defoliation. Despite this body of work on how insect outbreaks affect fire activity, there is a lack of consensus on how the spatial legacies of an eastern SBW outbreak might affect fire behaviour and containment in Ontario.

Despite the ecological benefits of wildfires, active fire suppression is always required in instances involving threats to people or values, especially in the wildland–urban interface. A primary goal of fire suppression is to contain the wildfire quickly to minimise the area burned and loss of values. Containment within Canada occurs when the Incident Commander classifies a fire as ‘being held’ (referred to in Ontario as BHE), which is an operational status assigned to a fire when it is considered unlikely to spread in forecast conditions owing to sufficient suppression resources and actions (Canadian Interagency Forest Fire Centre 2021). Wildfires that escape containment efforts can become very large and damaging, affecting both people and ecosystems (Fried *et al.* 2007). The probability that a fire will escape IA efforts is influenced by many factors, many of which are also associated with high fire intensity and increased rates of spread (Collins *et al.* 2018). The probability of fire containment can be affected by weather (Arienti *et al.* 2006; Beverly 2017; Cardil *et al.* 2018; Collins *et al.* 2018), fuel type (Flannigan *et al.* 2006; Cardil *et al.* 2018), fuel load (Collins *et al.* 2018), topography (Collins *et al.* 2018), fire behaviour (Cardil *et al.* 2018), agency response times (Arienti *et al.* 2006), allocation of resources (Flannigan *et al.* 2006; Collins *et al.* 2018), fire size at the time of IA (Arienti *et al.* 2006; Beverly 2017; Marshall *et al.* 2022) and time since prior wildfire disturbance (Beverly 2017). As fire spread rates increase and the fire perimeter grows, crews must establish a longer hose lay to contain the fire. Establishing a perimeter takes time, and if the fire increases in intensity, ground crews may be unable to control the fire (Collins *et al.* 2018). It is expected that the probability of containment will decrease owing to the cumulative effects of SBW defoliation on surface fuel accumulation, fire intensity and crowning potential (Stocks 1987; James *et al.* 2017; Candau *et al.* 2018; Watt *et al.* 2018), all of which can contribute to fires that are challenging to control (Beverly 2017; Collins *et al.* 2018).

In this study, we investigated the determinants of IA success in the boreal forests of Ontario, Canada, and whether the inclusion of SBW defoliation history improves predictive models of containment. Specifically, we developed random forest models of wildfire containment using historical fire occurrence records and aerially mapped SBW defoliation data. We compared models of IA success with and without information on SBW defoliation using several metrics of model performance, including accuracy, specificity, sensitivity and the area under the receiver operator curve (AUC). We hypothesised that SBW defoliation data would improve model performance owing to its effect on fuels. We

further hypothesised that defoliation from 6 to 9 years prior to the fire event (i.e. within the historical window of opportunity) would have a greater influence on IA. Understanding this relationship between insect-altered fuels and wildfire containment is important, as climate change is altering disturbance regimes in terms of their frequency, severity and extent, with potential disruption to ecosystem services (Gauthier *et al.* 2015). Fires that escape IA efforts can lead to widespread consequences to human health and the loss of values. With the additional uncertainty that comes with novel fuel types, ecosystems could shift to alternative stable states, thus disrupting numerous ecosystem services (Dale *et al.* 2001; Buma 2015; Walker *et al.* 2023).

Methods

Study region

We modelled IA success in the boreal forest region of Ontario, Canada (Fleming *et al.* 2002; Crins *et al.* 2009), a region with a long history of SBW defoliation (Fig. 1a) and wildfire disturbances (Fig. 1b). Ontario is divided into ecoregions based on similarities in climate, vegetation and bed-rock geology (Crins *et al.* 2009). Eastern ecoregions within the province have greater mean annual precipitation compared with the west (Crins *et al.* 2009). Western ecoregions typically experience more severe drought conditions that are conducive to frequent, large fires (Crins *et al.* 2009; Parisien *et al.* 2011). Forest types across the eastern and western ecoregions include a combination of Boreal and Great Lakes–Saint Lawrence forest types. Boreal forest fire regimes involve small frequent fires, with occasional large stand-replacing fires (Weir *et al.* 2000). In the mixed-wood forests, fire return intervals range between 63 and 210 years (Crins *et al.* 2009). As coniferous composition increases, the fire return interval decreases. Upland coniferous forests in these regions see fire return intervals of 30–187 years, and fires in jack pine stands range from 50 to 187 years, and are often stand replacing. Lowland coniferous forests have longer return intervals of 150–6000 years and are of varying intensity (Crins *et al.* 2009).

Data

We modelled a binary response of IA success using environmental and fire management predictors that have been previously shown to affect containment probability (Table 1). Environmental predictors such as fire weather, fuel type, SBW defoliation history, ecoregion and fire season have been shown to influence fire behaviour and thus have potential to create conditions in which a fire would be difficult to control. We also included fire management predictors such as fire size at onset of IA and fire crew response time. Each variable was spatially and temporally restricted to each fire ignition point included in this study.

Table 1. Variables that were included in the SBW and null random forest models.

Variable	Description
Size at IA	The size of the fire in hectares at the onset of initial attack efforts
Response time	The time between the time the fire was reported and the time of initial attack in hours
BUI	Build Up Index, a unitless indicator of the fuel available for consumption by a fire
ISI	Initial Spread Index, a unitless value representing potential fire spread
FFMC	Fine Fuel Moisture Code, a value that represents fuel moisture content of fine fuels and litter
DC	Drought Code, a value that represents moisture content in deep organic layers
FWI	Fire Weather Index, a unitless value representing fire intensity
DMC	Duff Moisture Code, a value that represents the moisture content in the organic layer
TSD	The time since the onset of defoliation in years before the fire ignition
Fuel type	The type of fuel the fire ignited in, as represented by the Fire Behaviour Prediction System fuel categories
Season	The season that the fire started in, a binary variable for spring and summer with a threshold of 15 June
Ecoregion	The Ontario ecoregion in which the fire ignited

IA success

A fire is classified as successfully contained when it is not expected to grow further owing to adequate resources and suppression efforts (i.e. BHE stage of control). There are many ways that a wildfire can be classified as contained or escaped, but for the purposes of this study, a fire was deemed contained if it was being held by 13:00 hours the day following IA. Previous work has found no significant difference in the classification of IA success and failure across definitions in modelling wildfire containment (Korkola *et al.* 2024). According to the definition of 13:00 hours to BHE, 83.1% of fires in Ontario are classified as successfully contained. Using unbalanced datasets for machine learning models can bias results towards the majority class. In this case, the model can classify all fires as contained and still have high overall accuracy. To correct for imbalance in the data, a down-sampling approach was used on the majority class during random forest building. Observations from majority and minority classes ($N = 700$) in the training data were randomly sampled with replacement for each iteration of the random forest.

Fire data

This study uses historical fire ignition data (1990–2019) for Ontario, Canada, provided by the Ontario Ministry of Natural Resources (OMNR). For each ignition, information documenting fire weather, fuel type, agency response time and fire season was extracted. Fire season was represented as a binary categorical variable such that ignitions before 15 June were considered spring fires, and ignitions after this date were considered summer fires (James *et al.* 2017). Agency response time was calculated by taking the difference between the reported time of the fire and the time of IA, represented in hours.

SBW defoliation data

Time since the onset of defoliation (TSD), reported in years from 0 to a maximum of 15 years, was used to include SBW defoliation in the model. This categorical SBW defoliation predictor was computed using aerially mapped SBW defoliation data (1975–2018) provided by the OMNR (James *et al.* 2011, 2017). Ignition points were joined to defoliation polygons using a buffer of 500 m around each ignition to account for spatial uncertainty in defoliation polygon boundaries. Other fuel types in our study were represented by categories in the Canadian Forest Fire Behaviour Prediction (FBP) System (Forestry Canada Fire Danger Group 1992). FBP System fuel type information for each ignition location was included in the OMNR ignition database. Fuel types included Spruce–Lichen Woodland (C-1), Boreal Spruce (C-2), Mature Jack Pine (C-3), Immature Jack Pine (C-4), Red and White Pine (C-5), Conifer Plantation (C-6), Boreal Mixedwood (M-1, M-2), Dead Balsam Fir Mixedwood (M-3, M-4), Deciduous (D), Slash (S), Matted Grass (O-1A) and Standing Grass (O-1B).

Weather data

Daily observations at 13:00 hours Local Daylight Time (LDT) for temperature, 24-h precipitation, relative humidity and wind are used as inputs for the Canadian Forest Fire Weather Index (FWI) System (Van Wagner 1974). The FWI System provides a relative rating of fuel moisture in three forest floor layers (Van Wagner 1974), represented by the Fine Fuel Moisture Code (FFMC), the Duff Moisture Code (DMC) and the Drought Code (DC). The DMC and DC together form the Build-up Index (BUI), which captures the amount of fuel available for combustion and the difficulty in fire extinguishment (Van Wagner 1974). The Initial Spread Index (ISI) describes the potential rate of spread of a fire based on FFMC and wind speed (Van Wagner 1974). Finally,

the ISI and BUI are used to calculate the FWI, which describes potential fire intensity (Van Wagner 1974). Fire weather indices were interpolated to each fire ignition location using the OMNR weather stations and thin plate splines (Flannigan and Wotton 1989; Wheatley *et al.* 2022).

Modelling

Random forest

The random forest procedure is a machine learning method that combines classification trees to identify more accurate and stable results than one would find using single trees (Breiman 2001). Combinations of independent variables are used to explain variation in the categorical response variable by continually splitting the data into more uniform groupings (De'Ath and Fabricius 2000). These splits aim to make the groups as homogeneous as possible based on the data's proportion of IA successes and failures. Prior studies have opted to apply random forest methods in IA success modelling owing to the high predictive accuracy and ability to avoid overfitting (Collins *et al.* 2018; Marshall *et al.* 2022). Using the *randomForest* package in R (Breiman 2001; Liaw and Wiener 2002), we created two random forest models, one with and one without TSD as a predictor. We used r times k -fold cross-validation using the *caret* package (Kuhn 2008), a technique in which the data are split into k number of equal folds. Cross-validation allows the testing of model performance on unseen data. In r times k -fold cross-validation, the data are trained on the $k-1$ fold while the other folds are used for testing. The process is repeated r times using different splits into k folds. The parameters for our cross-validation were $r = 10$ and $k = 10$.

Variable selection and importance

We used the mean decrease in accuracy (MDA) to determine the main drivers of wildfire containment. The MDA is a measure of how much accuracy the model loses by excluding that variable. Variable importance measures provide information on the main contributors to IA success. Low MDA values (i.e. values approaching zero) contribute very little to the model's overall accuracy whereas negative MDA values suggest predictive accuracy generally increases when these variables are removed. Model covariates with the lowest variable importance scores were iteratively removed until model accuracy was no longer improved. Removal of low-importance variables can increase model interpretability and redundancy across predictors in order to determine the main drivers of containment in Ontario. Partial dependence plots (PDPs) were used as a graphical representation of the marginal effects of each predictor on the response, in this case, the probability of containment. We used the *iml* package in R for all PDPs (Molnar *et al.* 2018).

We identified fire size at IA as the most important variable in both the null and SBW models (see Results) and therefore created a simple random forest model of fire size

at IA to determine whether TSD contributed to fire size. We used fire weather (FFMC, DMC, DC, ISI, BUI and FWI), response time, fuel type, season and TSD to model fire size at IA. We assessed variable importance using the percentage increase in mean squared error (%incMSE), which evaluates model accuracy when variables are permuted. Higher values represent greater importance in the model.

Analysis

Performance metrics

All statistical tests and analysis were completed using R statistical software (R Core Team 2024). The performance of our two models were evaluated using four commonly used machine learning metrics: overall accuracy, sensitivity, specificity and AUC using packages *caret* (Kuhn 2008) and *pROC* (Robin *et al.* 2011). Overall accuracy captures the model's ability to correctly classify a fire as escaped or contained out of all the predictions. Sensitivity refers to the model's ability to correctly classify IA success whereas specificity refers to the model's ability to classify IA failure correctly. Finally, AUC captures the model's ability to distinguish between successes and failures, where 1 is perfect distinguishment, and 0.5 is no better than random. Accuracy, sensitivity and specificity are all expressed on a continuous scale between 0 and 1, where 0 indicates poor distinguishment between classes and 1 indicates perfect discriminatory ability. The Youden Index was used to determine the optimal cut-off threshold for classification on the receiver operator curve (ROC) (Youden 1950). Using multiple metrics can help determine whether the model is overfitting or accurately predicting IA success from failure. The distribution of each performance metric generated from the repeated 10-fold cross-validation procedure was used in a modified paired t -test to determine if model metrics were significantly different.

Significance testing

We used a modified paired t -test at significance level $\alpha = 0.05$ (Nadeau and Bengio 1999; Bouckaert and Frank 2004) to determine if there were significant differences between the performance metrics derived from models that did and did not include TSD as a predictor. The t -test pairs were the outputs from the r times k -fold CV (Cross-Validation), in which there were 100 pairs for each accuracy metric. We did not use a classic t -test to compare the means because values taken from the output of a cross-validation violate the assumption of independence and often result in a high Type I error (Nadeau and Bengio 1999). The modified paired t -test was performed on the results of the repeated k -fold cross-validation test and is expressed as:

$$t = \frac{\frac{1}{kr} \sum_{i=1}^k \sum_{j=1}^r x_{ij}}{\sqrt{\left(\frac{1}{kr} + \frac{n_2}{n_1}\right) \hat{\sigma}^2}}$$

where k is the number of folds, r is the number of repetitions, x is the difference between the two means for run i in fold j , n_2 is the number of fires used for testing, and n_1 is the number of fires used for training.

Results

From the original 36,031 fires in the OMNR database, 5080 fires were retained for modelling after exclusions. Of these fires, 83.1% were considered a successful IA containment, and 55% occurred in fuels with a defoliation history of up to 15 years. Fire size at IA for contained fires averaged 0.63 ha, with a median value of 0.3 ha whereas the average initial fire size for escaped fires was 7.16 ha, with a median value of 1.5 ha (Fig. 2). We restricted our analysis to fires with a reported IA or BHE date. Fires lacking this information were assumed to be 'being observed', where suppression was not the main objective. Further restrictions included only modelling lightning-caused fires occurring between April and October. We restricted our analysis to lightning ignitions only, because anthropogenic fires tend to have different environmental drivers and are not equally distributed geographically. Fires outside the April–October fire season were excluded to limit confounding factors such as a reduction in available suppression resources. Fires with a final reported size of 0.1 ha were also excluded. In Ontario, the smallest reported fire size is 0.1 ha, but this category includes a range from 0.01 to 0.1 ha, making it challenging to accurately capture fire growth at these smaller sizes. Fires were also excluded if the response time exceeded 48 h, as these fires would have already escaped IA efforts and could be classified as response failures rather than containment failures ($n = 1124$).

The initial model of IA success included variables such as fire size at IA, fire weather indices (FFMC, DMC, DC, ISI,

BUI, FWI), ecoregion, fuel type, season, response time and TSD (time since onset of defoliation). The overall accuracy of the full model was 0.85 (s.d. 0.02) with an AUC of 0.81 (s.d. 0.03). Variables with low importance based on the MDA were iteratively removed until model accuracy was no longer improved.

Variables retained in the final SBW model included fire size at IA, FFMC, DMC, DC, BUI, ISI, FWI, ecoregion and TSD. The most important variable was the size of the fire at IA, followed by FWI and ISI (Fig. 3a). Response time, fuel type and season were removed owing to low MDA values.

Partial dependence plots (PDPs) were used to illustrate the marginal effects of each predictor on the probability of IA success with all other predictors held constant (Fig. 4). For the SBW model, the probability of containment decreased as the size at IA increased. When fire size at IA was 0.1 ha, the likelihood of containment was 92%. At 1.4 ha, there was a 50% chance of BHE by 13:00 hours the next day. Once fires surpassed 2 ha in size, the probability of containment dropped to 11%. Generally, as FWI values increased, the probability of containment steadily decreased. When ISI values were below 5, containment probability was generally higher, after which containment likelihood dropped and then steadily increased. Low BUI values (< 25) were associated with lower IA success probabilities, whereas higher values were typically associated with greater containment. As DMC increased, the probability of containment also increased, with the lowest IA success probability occurring at $DMC < 10$. Containment probability generally increased with higher FFMC, but decreased once values exceeded approximately 85. Although it was least important in the model, the PDPs for TSD showed that increasing TSD generally lowers the probability of containment, especially at time lags greater than 9 years from the onset of defoliation. A value of 0 represented fires that did not burn in defoliated

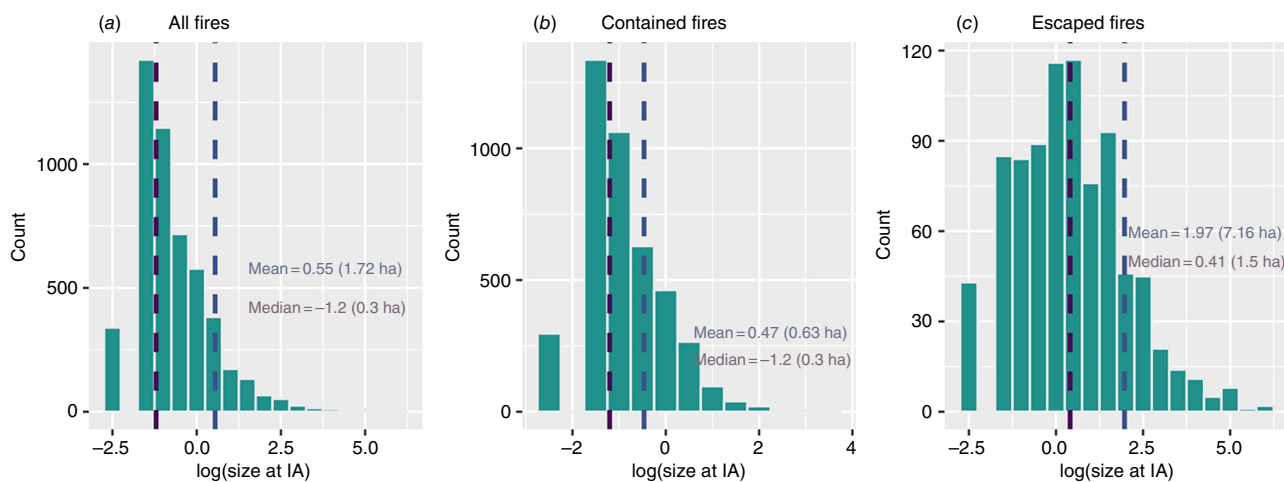


Fig. 2. Distribution of fire sizes at initial attack (IA). Values are presented as the natural logarithm. (a) The distribution for all fires for which the mean was 1.72 ha and the median was 0.3 ha. (b) All contained fires with average initial fire size of 0.63 ha with a median value of 0.3 ha. (c) Distribution for all escaped fires. The average size at IA for escaped fires was 7.16 ha, with a median value of 1.5 ha.

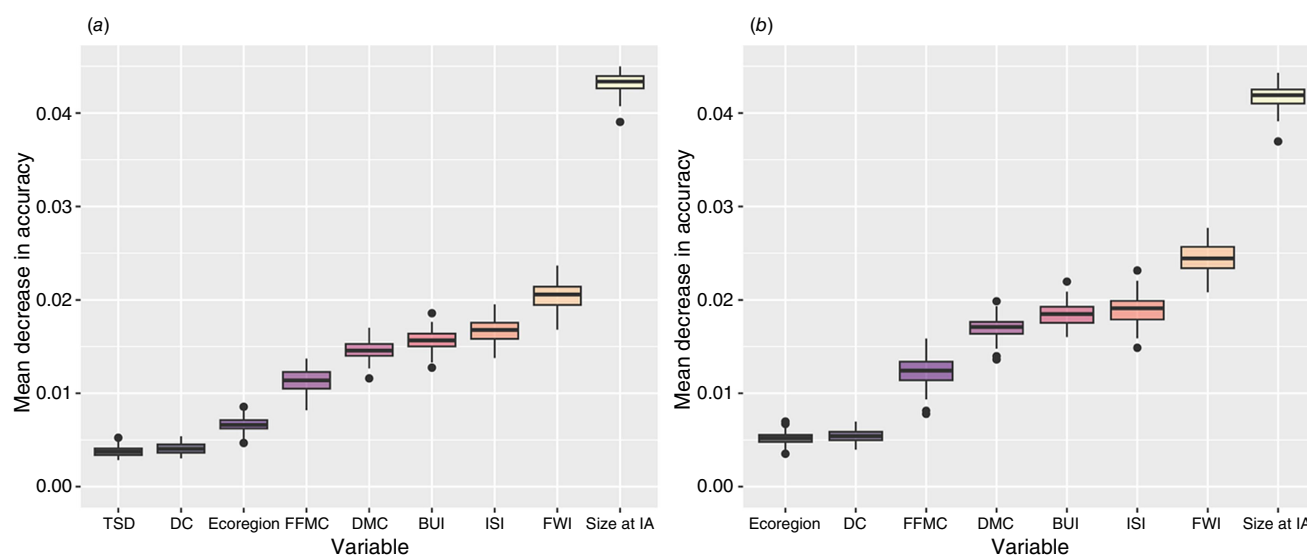


Fig. 3. Comparison of variable importance represented by the mean decrease in accuracy (MDA) for the (a) SBW model, and the (b) null model. Variables are ordered along the x axis in increasing order of the median MDA value. Fire Size at IA was the most important variable in both models. Abbreviations used are: TSD, time since onset of defoliation; DC, Drought Code; FFMC, Fine Fuel Moisture Code; DMC, Duff Moisture Code; BUI, Build-Up Index; ISI, Initial Spread Index; FWI, Fire Weather Index; IA, Initial Attack.

areas and showed a lower probability of containment compared with defoliated areas.

Model predictive accuracy was assessed using overall accuracy, specificity, sensitivity and AUC. The SBW model performed well, with values for overall accuracy, sensitivity and AUC approaching 1, representing a near-perfect class distinction between IA success and failure. Model specificity was generally lower owing to IA failures being rare events in our data. The out of bag (OOB) error was 17.78%. The overall mean accuracy of the SBW model was 0.852 (s.d. 0.016), AUC was 0.814 (s.d. 0.024), specificity was 0.507 (s.d. 0.056) and sensitivity was 0.922 (s.d. 0.014). Together, these metrics indicate good model performance.

A null model was also created that excluded SBW defoliation as a predictor. The final null model included size at IA, FWI, ISI, BUI, DMC, FFMC, DC and ecoregion. Response time, fuel type and season, were removed owing to low variable importance. The size of the fire at IA was again the most important predictor, followed by FWI and ISI. Of the lowest importance in the model were DC and ecoregion (Fig. 3b).

The fire size at IA had the highest MDA value in the null model. Variables of the highest importance in the model tend to show greater variation in the PDPs (Fig. 5). As size at IA increased, the probability that a fire was contained decreased sharply. At an initial fire size of 0.1 ha, the likelihood of containment was 92%. Once fires exceed 2 ha, the probability of containment dropped to less than 10%. As FWI increased, the probability of containment decreased, with high and extreme values having lower success rates. ISI values of ~5–7 tended to have lower containment. Lower BUI showed lower containment whereas higher values showed high containment likelihoods. Lower DMC

values had lower containment, but at values of ~10, containment likelihood increased.

The null model generally performed well across all accuracy metrics (Table 2) and was similar in terms of performance to the model that included SBW defoliation. The mean OOB error for this model was 18.15%. The mean overall accuracy was 0.852 (s.d. 0.017), the mean AUC was 0.806 (s.d. 0.025), the mean specificity was 0.515 (s.d. 0.058) and the model sensitivity was 0.921 (s.d. 0.015).

Paired *t*-tests were completed for each of the accuracy metrics: overall accuracy, AUC, sensitivity and specificity to assess for significant differences between the SBW and null model (Table 3). Although accuracy metric values differed slightly, these differences were generally not statistically significant (Table 3). Accuracy values were not significantly different ($t = -0.063$; d.f. = 99; $P = 0.950$), nor were specificity values ($t = -0.736$; d.f. = 99; $P = 0.463$) or sensitivity values ($t = 0.427$; d.f. = 99; $P = 0.670$). However, the difference between the SBW and null model AUC values was significantly different ($t = 2.798$; d.f. = 99; $P = 0.006$). The higher AUC value in the SBW model indicated that including SBW defoliation as a predictor improves the model's ability to distinguish between IA success and failure.

The size of the fire at IA was the most important determinant of IA success in both final models. Given our interest in the role of historical defoliation on containment success, we created a simple random forest model to determine if TSD influenced the fire size at IA. We found that the overall percentage variance explained was low (–1.52%), suggesting the model performs poorly and that size at IA is not influenced by the TSD predictor. We used the %incMSE to assess variable importance. Weather variables were the greatest predictor of

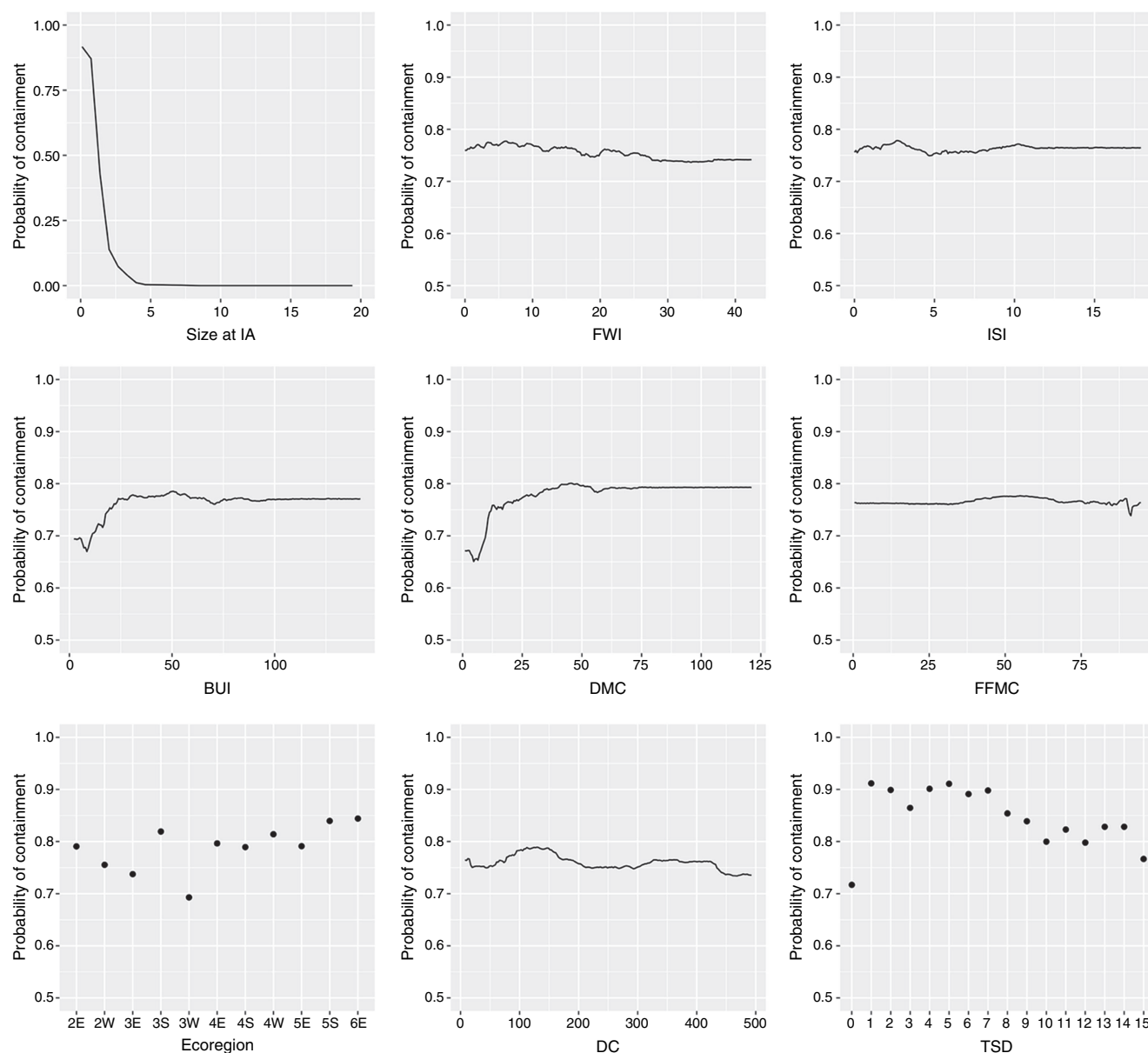


Fig. 4. Partial dependence plots (PDPs) for the SBW random forest model of IA success. PDPs are a graphical representation of single feature effects on the response variable. Predictors are shown by decreasing variable importance. Abbreviations used are: DC, Drought Code; FFMC, Fine Fuel Moisture Code; DMC, Duff Moisture Code; BUI, Build-Up Index; ISI, Initial Spread Index; FWI, Fire Weather Index; IA, Initial Attack; TSD, time since onset of defoliation.

size at IA, specifically FWI, DC and DMC. Response time was the next most important variable in the model. Fuels and season were of lower importance and TSD was of the lowest importance in the model. The variable importance plot is shown in Supplementary Material S1.

Discussion

This study modelled IA success probability in Ontario, Canada, using several environmental and fire management

variables, including spruce budworm defoliation history (TSD). Our objective was to evaluate whether the inclusion of TSD enhanced the predictive performance of IA models. Previous research has established a link between SBW defoliation and wildfire behaviour due to changes in forest fuel structure (Stocks 1987; Watt 2014; Watt *et al.* 2020). However, our findings suggest that the effect of SBW defoliation on IA success is limited.

Models that both did and did not include SBW defoliation history showed that the most important variables used to determine IA success were fire size at IA, FWI and ISI,

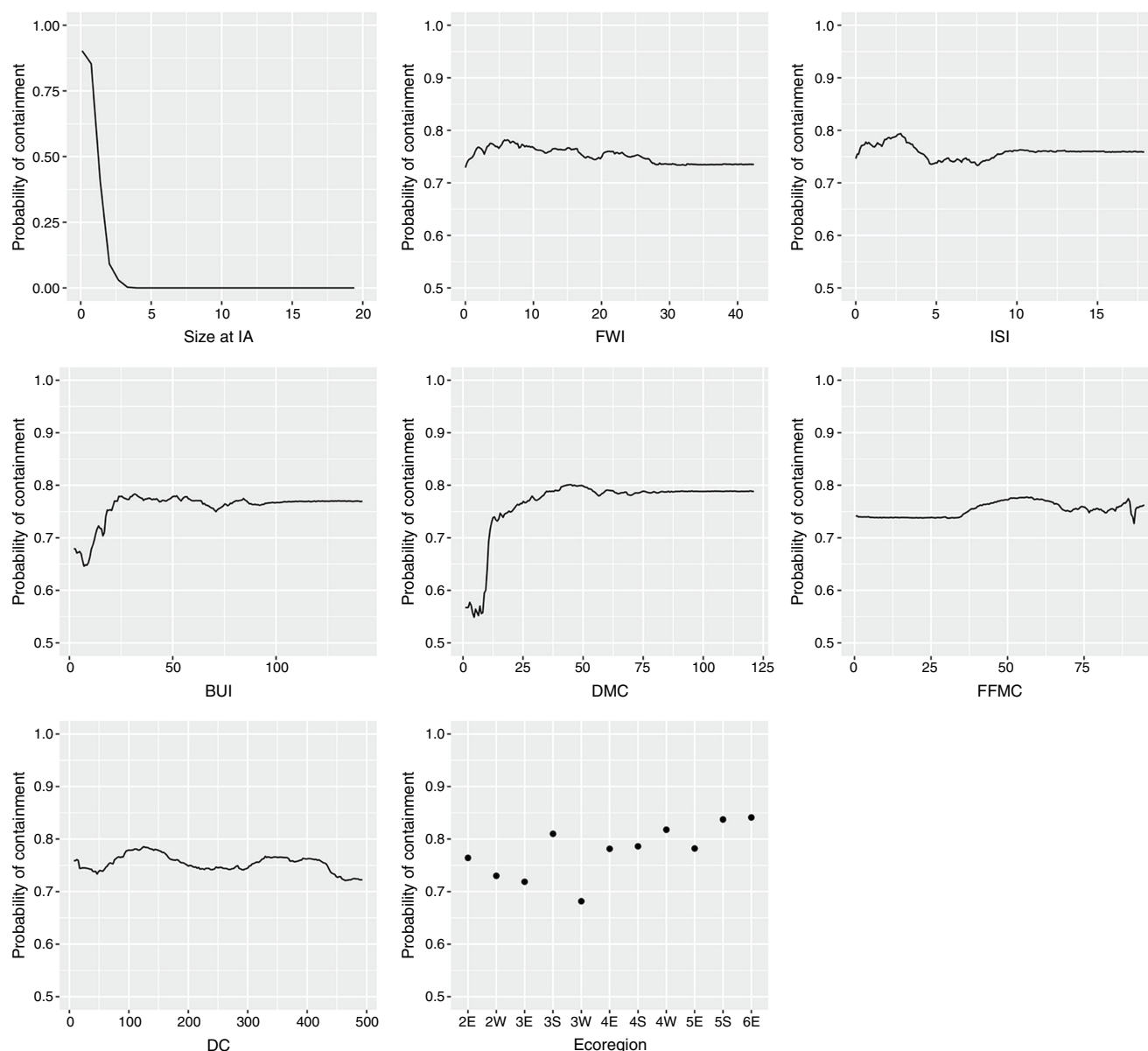


Fig. 5. Partial dependence plots for the null random forest model of IA success. Abbreviations used are: DC, Drought Code; FFMC, Fine Fuel Moisture Code; DMC, Duff Moisture Code; BUI, Build-Up Index; ISI, Initial Spread Index; FWI, Fire Weather Index; IA, Initial Attack.

consistent with previous findings (Arienti *et al.* 2006; Beverly 2017; Marshall *et al.* 2022). These factors reflect the difficulty in containing larger, more intense and rapidly spreading fires. As the fire size at IA increased, the probability of containment dropped rapidly, consistent with past research showing that IA is most successful on small fires (Reimer *et al.* 2019). Fires that surpassed 2 ha in size before IA had a less than 10% chance of successful containment by 13:00 hours the following day. Larger fires at the onset of IA typically demand more resources and occur under more extreme fire weather conditions, making direct suppression tactics more difficult and time consuming.

Other important variables in our models included the fire weather indices, mainly FWI and ISI. These predictors had lower MDA values compared with the fire size at IA; however, previous work has also shown these variables to be important for understanding IA success and fire behaviour (Arienti *et al.* 2006; Flannigan *et al.* 2006; Podur and Wotton 2010; Beverly 2017; Cardil *et al.* 2018; Collins *et al.* 2018). The FWI is a general indicator of fire intensity, in which higher values reflect conditions under which fires are more difficult to suppress. The ISI, an index of rate of fire spread, also showed an inverse relationship with IA success. However, these relationships were more complex, as the

Table 2. The mean and s.d. for the random forest performance metrics.

Metric	Mean $\pm$ s.d.	
	SBW model	Null model
Accuracy	0.852 $\pm$ 0.016	0.852 $\pm$ 0.017
AUC	0.814 $\pm$ 0.024	0.806 $\pm$ 0.025
Specificity	0.507 $\pm$ 0.056	0.515 $\pm$ 0.058
Sensitivity	0.922 $\pm$ 0.14	0.921 $\pm$ 0.015
OOB error (%)	17.78	18.15

Means and s.d. were a result of the repeated k fold cross validation ($r = 10$, $k = 10$). The SBW model includes time since onset of defoliation (TSD) as a predictor whereas the null model does not. Abbreviations used are: SBW, spruce budworm; AUC, area under the receiver operator curve; OOB, out of bag.

Table 3. The t -scores and P -values for each model were calculated from the modified paired t -test (d.f. = 99, $\alpha = 0.05$) for each performance metric.

Metric	t -score	P -value
Accuracy	-0.063	0.950
AUC	2.798	0.006
Specificity	-0.736	0.463
Sensitivity	0.427	0.670

A paired t -test was used to determine if there were significant statistical differences between the means of the SBW and the null model. AUC was the only performance metric with significant differences ($\alpha < 0.05$). Abbreviations used are: SBW, spruce budworm; AUC, area under the receiver operator curve.

partial dependence plots suggest that containment likelihood first decreases and then increases as ISI values rose above 5. These non-linear effects likely reflect the dynamic interaction between fire behaviour, resource availability and suppression tactics. Fire management agencies can adjust their tactics and prioritise fires when fire weather indices are high to ensure quick response and sufficient resources. However, we also acknowledge that the fire weather indices are often highly correlated (see Supplementary Material S2). Random forest models use ensemble learning and a random subset of predictors at each split, making them less affected by this problem. As a result, we chose to include all the FWI predictors, even though FWI and BUI were highly correlated with the other fire weather indices (Supplementary Material S2). However, variable importance scores (MDA) can be affected by correlation among predictors. To further determine if our results were sensitive to high correlation between the fire weather indices, we built the same random forest models (SBW and null model) but without the inclusion of BUI and FWI. We found that our main results were robust and that size at IA was still the most important variable followed by ISI and FFMCI. The variable importance plots can be found in Supplementary Material S3.

The inclusion of spruce budworm defoliation history through the TSD variable did not considerably improve model predictive performance, despite the previously described effects of SBW on forest fuels and fire behaviour (Stocks 1987; Fleming et al. 2002; James et al. 2017; Candau et al. 2018; Watt et al. 2018, 2020; Fettig et al. 2022). However, there was evidence that AUC was slightly, and significantly, higher in the model that included TSD, suggesting that the SBW model was better able to distinguish between IA success and failure. The PDPs indicate that areas with long-term defoliation (> 9 years) may have marginally reduced IA success, which may reflect structural changes in fuels (Watt et al. 2018, 2020) and associated hypothesised increases in fire intensity (Stocks 1987). Previous work has identified a 'window of opportunity', typically 3–9 years following an outbreak, during which fire risk and fire behaviour are highest (Stocks 1987; Fleming et al. 2002; James et al. 2017). This period is characterised by the accumulation of ladder fuels, which enhance the potential for surface fires to transition into crown fires by creating a more continuous vertical fuel structure (Stocks 1987; Watt et al. 2018). Increased crown breakage and the falling of standing dead trees contribute to an increase in the surface fuel load and eventual reduction in canopy fuels (Flower et al. 2014; Watt et al. 2020). Our findings indicate that although SBW defoliation may improve model discriminatory ability, its influence on the probability of successful IA appears to be limited compared with factors like fire size and weather.

Possible reasons for the relatively small effect of SBW on IA success in this study could be related to the scale and timing of defoliation data used. We used spatially aggregated, provincial-level defoliation data that may not capture the fine-scale variations in fuel conditions that could influence the effectiveness of IA operations. FBP fuel types may not accurately capture stand structural changes from defoliation that are relevant for understanding IA or fire behaviour in these altered stands (Phelps and Beverly 2022). Additionally, we did not account for stand age or succession in our fuel representation; instead, we used a time since onset of defoliation variable (TSD), which may not capture fine-scale fuel changes or follow predictable succession pathways (Beverly 2017). Much of the work investigating the interaction between insect-related fuel changes and wildfire is that of statistical modelling and simulation (Lynch and Moorcroft 2008; James et al. 2017; Candau et al. 2018) rather than empirical field studies (Watt et al. 2018; Watt et al. 2020). Future research should focus on better quantifying stand-scale insect-related fuel changes and how these changes influence wildfire behaviour directly rather than through human thresholds of suppression. Alternatively, the impact of SBW defoliation may be more pronounced during moderate fire weather, where additional fuel loads could amplify fire behaviour and overwhelm suppression capabilities. Many fires that escaped IA in the present study occurred during extreme fire weather and

with large IA sizes, a scenario in which the added complexity of altered fuels may be masked (Hart *et al.* 2015).

Other possible reasons for the lack of SBW defoliation influence on IA success could also be increased crown loss after top-kill, which reduces the canopy fuel load and thus the potential for torching and crown fire spread (Stocks 1987; Lynch and Moorcroft 2008; Flower *et al.* 2014). Other factors could be related to fuel decomposition rates and understorey proliferation, which would reduce the surface fuel load and increase humidity within the stand (Péché 1993; James *et al.* 2017). Regardless, this result showing the lack of an effect of SBW on IA is important as it highlights the complex relationships between insect outbreaks and wildfires (Fettig *et al.* 2022). Although some aspects of wildfire activity may be affected by outbreak legacies (e.g. ignition; James *et al.* 2017), it is not necessarily so that other aspects (e.g. IA success) are equally affected. More research is needed to better understand the complex, spatially varying relationships between insect outbreak impacts, fire behaviour and the efficacy of IA suppression.

We further explored the effect of fuels and TSD on the size of the fire at IA. We reasoned that if TSD did not directly affect IA, and that size at IA was the most important determinant, perhaps TSD indirectly affected IA success through its influence on the size of a fire at IA. Defoliation alters the fuel complex, potentially increasing the surface fuel load, increasing fuel drying from greater canopy openness, and promoting greater connectivity between surface and crown fuels. These insect-related changes can alter fire intensity and rates of spread, thus contributing to greater area burned. However, we found that weather and response time were the main determinants of initial fire size, whereas fuels and TSD were of low importance. We understand that modelling fire size at this spatial and temporal scale could lead to the lack of influence of fuels on fire size, in which large amounts of variation are not accounted for by weather, response times and fuels alone. This complex relationship requires further study that is beyond the scope of this paper.

For this study, we defined a fire as contained when it was BHE before 13:00 hours the day after IA. This period is critical during suppression, as fire agencies want to contain a fire before the following burning period, in which fire behaviour peaks and therefore fires become much more difficult to control. This definition was chosen to limit the conditions of the fuels and weather to the ignition points for which we had available data. We assumed that a fire could be deemed difficult within the first firefighting efforts as a result of the SBW fuels, which could delay crews from establishing the fire perimeter and thus deeming it BHE or contained. Most studies use either a time-based definition (Collins *et al.* 2018; Marshall *et al.* 2022; Wheatley *et al.* 2022), as we used here, or a size-based definition (Arienti *et al.* 2006; Podur and Wotton 2010). We found that a size definition was not appropriate for our study as we were only looking at the initial conditions surrounding fire ignition.

Also, previous work has found that the definition of successful containment does not greatly impact model results, and that these definitions should be chosen based on specific objectives (Korkola *et al.* 2024). The use of a threshold for IA success could also be a reason for the lack of evidence of SBW on containment probability rather than the use of fire behaviour outputs. Previous work has linked defoliation to fire behaviour (Fettig *et al.* 2022); however, this dynamic process in which fuels, weather and topography affect fire behaviour may not be well represented in an arbitrarily defined threshold such as fire containment.

The defoliator–wildfire interaction is challenging to quantify owing to the dynamic temporal and spatial changes in forest fuels, in which one must account for the effects of cumulative defoliation and the time since the defoliation or mortality. Improving our understanding of the mechanisms driving fire behaviour is crucial for enhancing our ability to model the complex interactions between insect outbreaks and wildfire dynamics. This lack of consensus on whether SBW increases fire behaviour or risk is in part due to the range of spatial and temporal scales that these disturbances occur at as well as the wide range of fire behaviour variables that can be used as a response (James *et al.* 2017; Fettig *et al.* 2022; Romualdi *et al.* 2023). This interaction between wildfire and defoliators also varies across its boreal range with evidence in the eastern parts of Canada showing no increase in wildfire risk following defoliation from the eastern SBW (Péché 1993). These results are limited to the eastern SBW and may not be applicable to other insect systems like the mountain pine beetle (*Dendroctonus ponderosae*) and western spruce budworm.

This study was restricted to lightning fires that occurred between 1990 and 2019 for the province of Ontario. We focused on lightning-caused fires to isolate the natural drivers of fire occurrence and minimise the influence of human ignitions. However, this focus on a specific ignition type could be a potential reason that season and response time had low importance in the initial models. Human-caused fires typically dominate in the spring, whereas lightning-caused fires dominate in the summer (Stocks *et al.* 2002). Restricting our analysis to lightning fires would reduce the variability in the season predictor and give it low importance in our model. Additionally, response times are generally longer for lightning ignitions compared with human-caused ignitions, which tend to be in more densely populated areas with quicker detection (Wotton and Martell 2005).

Conclusion

The boreal forest is expected to be the most affected by future climate change of all biomes (Gauthier *et al.* 2015). As a result, disturbance regimes are expected to shift in terms of their frequency, severity and extent. Climate change is expected to affect forest insect pests, including

the outbreak dynamics and spatial extent of the SBW, through both direct and indirect effects (Candau and Fleming 2011). Independently of changes to fuel abundances and connectivity due to insect activity, forest fires are also increasing in frequency and severity owing to climate change (Podur and Wotton 2010). Evidence suggests that the altered stand structure left behind by the SBW increases the probability of fire ignition (James et al. 2017) and severity (Fleming et al. 2002). These changes could also impede crew movement and alter air attack effectiveness, therefore affecting fire suppression efforts (Beverly 2017). We can expect to see a greater interaction between these two disturbances due to the considerable spatial extent of the SBW as well as the effects of climate change (Fleming et al. 2002; Candau and Fleming 2011; Régnière et al. 2012). These impacts will disrupt many ecosystem services, forest habitats and economic benefits.

Contrary to expectation, we did not find any strong evidence that SBW defoliation on its own affected wildfire containment probability. Rather, fire size at the time of initial attack and fire weather conditions remain the strongest predictors of IA success. Although SBW defoliation did not greatly improve model performance, it remains a variable of interest owing to its potential effects on fire ignition and spread. Future research should further investigate the initial fire behaviour in SBW-affected areas, especially considering the expected increase in fire load due to climate change. Understanding the dynamic changes in fuels due to defoliation and decomposition rates is needed for improving fire management strategies in the context of increasing wildfire activity and increasing uncertainty due to climate change.

Supplementary material

Supplementary material is available online.

References

- Arienti MC, Cumming SG, Boutin S (2006) Empirical models of forest fire initial attack success probabilities: the effects of fuels, anthropogenic linear features, fire weather, and management. *Canadian Journal of Forest Research* 36, 3155–3166. doi:10.1139/x06-188
- Beverly JL (2017) Time since prior wildfire affects subsequent fire containment in black spruce. *International Journal of Wildland Fire* 26, 919–929. doi:10.1071/WF17051
- Blais JR (1983) Trends in the frequency, extent, and severity of spruce budworm outbreaks in eastern Canada. *Canadian Journal of Forest Research* 13(4), 539–547. doi:10.1139/x83-079
- Bouckaert RR, Frank E (2004) Evaluating the replicability of significance tests for comparing learning algorithms. In 'Proceedings of Pacific-Asia Conference on Knowledge Discovery and Data Mining'. pp. 3–12. (Springer)
- Breiman L (2001) Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:1010933404324
- Buma B (2015) Disturbance interactions: characterization, prediction, and the potential for cascading effects. *Ecosphere* 6, 1–15. doi:10.1890/ES15-00058.1
- Canadian Interagency Forest Fire Centre (2022) Canadian Wildland Fire Glossary. Available at https://ciffc.ca/sites/default/files/2022-03/CWFM_glossary_EN.pdf
- Candau JN, Fleming RA (2011) Forecasting the response of spruce budworm defoliation to climate change in Ontario. *Canadian Journal of Forest Research* 41, 1948–1960. doi:10.1139/x11-134
- Candau JN, Fleming RA, Wang X (2018) Ecoregional patterns of spruce budworm–wildfire interactions in central Canada's forests. *Forests* 9, 137. doi:10.3390/f9030137
- Cardil A, Lorente M, Boucher D, Boucher J, Gauthier S (2018) Factors influencing fire suppression success in the province of Quebec (Canada). *Canadian Journal of Forest Research* 49, 531–542. doi:10.1139/cjfr-2018-0272
- Cohn GM, Parsons RA, Heyerdahl EK, Gavin DG, Flower A (2014) Simulated western spruce budworm defoliation reduces torching and crowning potential: a sensitivity analysis using a physics-based fire model. *International Journal of Wildland Fire* 23(5), 709–720. doi:10.1071/WF13074
- Collins KM, Price OF, Penman TD (2018) Suppression resource decisions are the dominant influence on containment of Australian forest and grass fires. *Journal of Environmental Management* 228, 373–382. doi:10.1016/j.jenvman.2018.09.031
- Countryman CM (1966) Rating fire danger by the multiple basic index system. *Journal of Forestry* 64(8), 531–536. doi:10.1093/jof/64.8.531
- Crins WJ, Gray PA, Uhlig PWC, Wester MC (2009) The ecosystems of Ontario, Part 1: Ecozones and ecoregions. Ontario Ministry of Natural Resources, Peterborough Ontario, Canada. Inventory, Monitoring and Assessment, SIB TER IMA TR-01. p. 71.
- Dale VH, Joyce LA, McNulty S, Neilson RP, Ayres MP, Flannigan MD, Hanson PJ, Irland LC, Lugo AE, Peterson CJ (2001) Climate change and forest disturbances: climate change can affect forests by altering the frequency, intensity, duration, and timing of fire, drought, introduced species, insect and pathogen outbreaks, hurricanes, windstorms, ice storms, or landslides. *BioScience* 51(9), 723–734. doi:10.1641/0006-3568(2001)051[0723:CCAFD]2.0.CO;2
- De'Ath G, Fabricius KE (2000) Classification and regression trees: a powerful yet simple technique for ecological data analysis. *Ecology* 81, 3178–3192. doi:10.1890/0012-9658(2000)081[3178:CARTAP]2.0.CO;2
- Fettig CJ, Runyon JB, Homicz CS, James P, Ulyshen MD (2022) Fire and insect interactions in North American forests. *Current Forestry Reports* 8, 301–316. doi:10.1007/s40725-022-00170-1
- Flannigan M, Wotton B (1989) A study of interpolation methods for forest fire danger rating in Canada. *Canadian Journal of Forest Research* 19(8), 1059–1066. doi:10.1139/x89-161
- Flannigan M, Amiro BD, Logan KA, Stocks BJ, Wotton BM (2006) Forest fires and climate change in the 21st century. *Mitigation and Adaptation Strategies for Global Change* 11, 847–859. doi:10.1007/s11027-005-9020-7
- Fleming RA, Candau JN, McAlpine RS (2002) Landscape-scale analysis of interactions between insect defoliation and forest fire in central Canada. *Climatic Change* 55, 251–272. doi:10.1023/A:1020299422491
- Flower A, Gavin D, Heyerdahl E, Parsons R, Cohn G (2014) Drought-triggered western spruce budworm outbreaks in the interior Pacific Northwest: a multi-century dendrochronological record. *Forest Ecology and Management* 324, 16–27. doi:10.1016/j.foreco.2014.03.042
- Forestry Canada Fire Danger Group (1992) Development and structure of the Canadian forest fire behavior prediction system. Information Report ST-X-3. (Forestry Canada, Science and Sustainable Development Directorate: Ottawa, ON)
- Fried JS, Gilles JK, Riley WJ, Moody TJ, Simon de Blas C, Hayhoe K, Moritz M, Stephens S, Torn M (2007) Predicting the effect of climate change on wildfire behavior and initial attack success. *Climatic Change* 87, 251–264. doi:10.1007/s10584-007-9360-2
- Gauthier S, Bernier P, Kuuluvainen T, Shvidenko AZ, Schepaschenko DG (2015) Boreal forest health and global change. *Science* 349(6250), 819–822. doi:10.1126/science.aaa9092
- Gray DR (2008) The relationship between climate and outbreak characteristics of the spruce budworm in eastern Canada. *Climatic Change* 87(3), 361–383. doi:10.1007/s10584-007-9317-5
- Hart SJ, Schoennagel T, Veblen TT, Chapman TB (2015) Area burned in the western United States is unaffected by recent mountain pine beetle outbreaks. *Proceedings of the National Academy of Sciences* 112(14), 4375–4380. doi:10.1073/pnas.1424037112

- James PM, Fortin M-J, Sturtevant B, Fall A, Kneeshaw D (2011) Modelling spatial interactions among fire, spruce budworm, and logging in the boreal forest. *Ecosystems* 14(1), 60–75. doi:10.1007/s10021-010-9395-5
- James P, Robert LE, Wotton BM, Martell DL, Fleming RA (2017) Lagged cumulative spruce budworm defoliation affects the risk of fire ignition in Ontario, Canada. *Ecological Applications* 27, 532–544. doi:10.1002/eap.1463
- Jardon Y, Morin H, Dutilleul P (2003) Périodicité et synchronisme des épidémies de la tordeuse des bourgeons de l'épinette au Québec. *Canadian Journal of Forest Research* 33(10), 1947–1961. [In French] doi:10.1139/x03-108
- Korkola K, Wheatley M, Beverly J, James PM, Wotton M (2024) A comparative analysis of wildfire initial attack containment objectives and modelling strategies in Ontario, Canada. *International Journal of Wildland Fire* 33(12), WF24104. doi:10.1071/WF24104
- Kuhn M (2008) Building predictive models in R using the caret package. *Journal of Statistical Software* 28, 1–26. doi:10.18637/jss.v028.i05
- Liaw A, Wiener M (2002) Classification and regression by randomForest. *R News* 2(3), 18–22.
- Lynch HJ, Moorcroft PR (2008) A spatiotemporal Ripley's K-function to analyze interactions between spruce budworm and fire in British Columbia, Canada. *Canadian Journal of Forest Research* 38(12), 3112–3119. doi:10.1139/X08-143
- MacLean DA (1980) Vulnerability of fir-spruce stands during uncontrolled spruce budworm outbreaks: a review and discussion. *The Forestry Chronicle* 56(5), 213–221. doi:10.5558/tfc56213-5
- MacLean DA (1984) Effects of spruce budworm outbreaks on the productivity and stability of balsam fir forests. *The Forestry Chronicle* 60(5), 273–279. doi:10.5558/tfc60273-5
- Marshall E, Dorph A, Holyland B, Filkov A, Penman TD (2022) Suppression resources and their influence on containment of forest fires in Victoria. *International Journal of Wildland Fire* 31(12), 1144–1154. doi:10.1071/WF22029
- Meigs GW, Campbell JL, Zald HS, Bailey JD, Shaw DC, Kennedy RE (2015) Does wildfire likelihood increase following insect outbreaks in conifer forests? *Ecosphere* 6(7), 1–24. doi:10.1890/ES15-00037.1
- Molnar C, Casalicchio G, Bischl B (2018) iml: an R package for interpretable machine learning. *Journal of Open Source Software* 3(26), 786. doi:10.21105/joss.00786
- Morris RF (1963) The dynamics of epidemic spruce budworm populations. *The Memoirs of the Entomological Society of Canada* 95(S31), 1–12. doi:10.4039/entm9531fv
- Nadeau C, Bengio Y (1999) Inference for the generalization error. In 'Advances in neural information processing systems. Vol. 12'. (MIT Press)
- Parisien M-A, Parks SA, Krawchuk MA, Flannigan MD, Bowman LM, Moritz MA (2011) Scale-dependent controls on the area burned in the boreal forest of Canada, 1980–2005. *Ecological Applications* 21(3), 789–805. doi:10.1890/10-0326.1
- Péché G (1993) Fire hazard in budworm-killed balsam fir stands on Cape Breton Highlands. *The Forestry Chronicle* 69(2), 178–186. doi:10.5558/tfc69178-2
- Phelps N, Beverly JL (2022) Classification of forest fuels in selected fire-prone ecosystems of Alberta, Canada—implications for crown fire behaviour prediction and fuel management. *Annals of Forest Science* 79(1), 40. doi:10.1186/s13595-022-01151-x
- Podur J, Wotton M (2010) Will climate change overwhelm fire management capacity? *Ecological Modelling* 221, 1301–1309.
- R Core Team (2024) 'R: A language and environment for statistical computing.' (R Foundation for Statistical Computing: Vienna, Austria) Available at <https://www.R-project.org/>
- Régnière J, St-Amant R, Duval P (2012) Predicting insect distributions under climate change from physiological responses: Spruce budworm as an example. *Biological Invasions* 14, 1571–1586. doi:10.1007/s10530-010-9918-1
- Reimer J, Thompson DK, Povak N (2019) Measuring initial attack suppression effectiveness through burn probability. *Fire* 2(4), 60. doi:10.3390/fire2040060
- Robin X, Turck N, Hainard A, Tiberti N, Lisacek F, Sanchez J-C, Müller M (2011) pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics* 12, 77. doi:10.1186/1471-2105-12-77
- Romualdi DC, Wilkinson SL, James P (2023) On the limited consensus of mountain pine beetle impacts on wildfire. *Landscape Ecology* 38(9), 2159–2178. doi:10.1007/s10980-023-01720-z
- Stocks B, Mason J, Todd J, Bosch E, Wotton B, Amiro B, Flannigan M, Hirsch K, Logan K, Martell D (2002) Large forest fires in Canada, 1959–1997. *Journal of Geophysical Research: Atmospheres* 107(D1), FFR 5-1–FFR 5-12. doi:10.1029/2001JD000484
- Stocks BJ (1987) Fire potential in the spruce budworm-damaged forests of Ontario. *The Forestry Chronicle* 63, 8–14. doi:10.5558/tfc63008-1
- Turner MG (1989) Landscape ecology: the effect of pattern on process. *Annual Review of Ecology and Systematics* 20, 171–197. doi:10.1146/annurev.es.20.110189.001131
- Turner MG (2005) Landscape ecology: what is the state of the science? *Annual Review of Ecology, Evolution, and Systematics* 36, 319–344. doi:10.1146/annurev.ecolsys.36.102003.152614
- Van Wagner CE (1974) 'Structure of the Canadian Forest Fire Weather Index.' (Environment Canada, Forestry Service Alberta, Canada)
- Wagner CV (1977) Conditions for the start and spread of crown fire. *Canadian Journal of Forest Research* 7(1), 23–34. doi:10.1139/x77-004
- Walker X, Okano K, Berner L, Massey R, Goetz S, Johnstone J, Mack M (2023) Shifts in ecological legacies support hysteresis of stand type conversions in boreal forests. *Ecosystems* 26(8), 1796–1805. doi:10.1007/s10021-023-00866-w
- Watt GA (2014) Enhanced vertical fuel continuity in forests defoliated by spruce budworm (*Choristoneura fumiferana* Clem.) promotes the transition of a surface fire into a crown fire. PhD Thesis, University of Toronto, Canada.
- Watt GA, Fleming RA, Smith SM, Fortin M-J (2018) Spruce budworm (*Choristoneura fumiferana* Clem.) defoliation promotes vertical fuel continuity in Ontario's boreal mixedwood forest. *Forests* 9(5), 256. doi:10.3390/f9050256
- Watt GA, Stocks BJ, Fleming RA, Smith SM (2020) Stand breakdown and surface fuel accumulation due to spruce budworm (*Choristoneura fumiferana*) defoliation in the boreal mixedwood forest of central Canada. *Canadian Journal of Forest Research* 50(6), 533–541. doi:10.1139/cjfr-2019-0076
- Weber MG, Stocks BJ (1998) Forest fires and sustainability in the boreal forests of Canada. *Ambio* 27(7), 545–550.
- Weir J, Johnson E, Miyanishi K (2000) Fire frequency and the spatial age mosaic of the mixed-wood boreal forest in western Canada. *Ecological Applications* 10(4), 1162–1177. doi:10.1890/1051-0761(2000)010[1162:FFATSA]2.0.CO;2
- Wheatley M, Wotton BM, Woolford DG, Martell DL, Johnston JM (2022) Modelling initial attack success on forest fires suppressed by air attack in the province of Ontario, Canada. *International Journal of Wildland Fire* 31(8), 774–785. doi:10.1071/WF22006
- Williams DW, Liebhold AM (2000) Spatial synchrony of spruce budworm outbreaks in eastern North America. *Ecology* 81(10), 2753–2766. doi:10.1890/0012-9658(2000)081[2753:SSOSBO]2.0.CO;2
- Wotton BM, Martell DL (2005) A lightning fire occurrence model for Ontario. *Canadian Journal of Forest Research* 35(6), 1389–1401. doi:10.1139/x05-071
- Youden WJ (1950) Index for rating diagnostic tests. *Cancer* 3(1), 32–35. doi:10.1002/1097-0142(1950)3:1<32::aid-cnrc2820030106>3.0.co;2-3

Data availability. The data used were provided by the OMNR and were used under licence.

Conflicts of interest. Jennifer Beverly is an Associate Editor of the *International Journal of Wildland Fire* but was not involved in the peer review or any decision-making process for this paper. The authors have no further conflicts of interest to declare.

Declaration of funding. We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada – Doctoral Canadian Graduate Scholarship to Kennedy Korkola, the OMNR through a Collaborative Research Agreement with The University of Toronto (PMAJ), and the NSERC/Canada Wildfire Strategic Network Grant to PMAJ.

Acknowledgements. We gratefully acknowledge data shared by the OMNR.

Author affiliations

^AInstitute of Forestry and Conservation, John H. Daniels Faculty of Architecture, Landscape and Design, University of Toronto, Toronto, ON M5S 3B3, Canada.

^BDepartment of Renewable Resources, 751 General Services Building, University of Alberta, Edmonton, AB, T6G 2H1, Canada.